EE3621: Digital Signal Processing

Lecture 8: Properties of the DFT & Circular Convolution

(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Motivation: Why study DFT properties? (FFT derivations and filtering applications).
- **05:00 – 15:00 (10 mins):** Fundamental properties: Periodicity, Linearity, and Conjugate Symmetry.
- **15:00 – 25:00 (10 mins):** Circular Shift of a sequence in time and frequency domains; comparison with linear shift.
- **25:00 – 35:00 (10 mins):** Circular Convolution – definition, properties, and comparison with linear convolution. Matrix methods.
- **35:00 – 40:00 (5 mins):** Parseval's Theorem, Review checkpoints, and Q&A.

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2 Fundamental Properties of the DFT

Let $x[n] \longleftrightarrow X[k]$ represent an N-point DFT pair, defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn} \tag{1}$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} \tag{2}$$

where $W_N = e^{-j\frac{2\pi}{N}}$ is the twiddle factor.

2.1 Periodicity

Theorem (Periodicity): If $x[n]$ and $X[k]$ form an N-point DFT pair, then they are implicitly periodic with period N:

$$X[k+N] = X[k] \tag{3}$$

$$x[n+N] = x[n] \tag{4}$$

Proof for Frequency Domain ($X[k+N] = X[k]$):

$$X[k+N] = \sum_{n=0}^{N-1} x[n] W_N^{(k+N)n} \tag{5}$$

$$= \sum_{n=0}^{N-1} x[n] W_N^{kn} W_N^{Nn} \tag{6}$$

Since $W_N = e^{-j\frac{2\pi}{N}}$, we have $W_N^{Nn} = \left(e^{-j\frac{2\pi}{N}}\right)^{Nn} = e^{-j2\pi n} = 1$. Substituting this back:

$$X[k+N] = \sum_{n=0}^{N-1} x[n] W_N^{kn}(1) = X[k] \tag{7}$$

Proof for Time Domain ($x[n+N] = x[n]$):

$$x[n+N] = \frac{1}{N}\sum_{k=0}^{N-1} X[k] W_N^{-k(n+N)} \tag{8}$$

$$= \frac{1}{N}\sum_{k=0}^{N-1} X[k] W_N^{-kn} W_N^{-kN} \tag{9}$$

Since $W_N^{-kN} = e^{j2\pi k} = 1$ for any integer k:

$$x[n+N] = \frac{1}{N}\sum_{k=0}^{N-1} X[k] W_N^{-kn} = x[n] \tag{10}$$

Engineering Intuition: Sampling in the frequency domain inherently causes a periodic extension in the time domain. A finite sequence is treated as one period of a periodic signal.

2.2 Conjugate Symmetry (Hermitian Symmetry)

Theorem (Symmetry for Real Signals): If $x[n]$ is real ($x[n] = x^*[n]$), then the DFT exhibits conjugate symmetry: $X[k] = X^*[N-k]$.

Proof:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} \tag{11}$$

Taking the complex conjugate:

$$X^*[k] = \sum_{n=0}^{N-1} x^*[n] e^{j\frac{2\pi}{N}kn} = \sum_{n=0}^{N-1} x[n] e^{j\frac{2\pi}{N}kn} \tag{12}$$

Now evaluate $X[N-k]$:

$$X[N-k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}(N-k)n} \tag{13}$$

$$= \sum_{n=0}^{N-1} x[n] e^{-j2\pi n} e^{j\frac{2\pi}{N}kn} \tag{14}$$

Since $e^{-j2\pi n} = 1$:

$$X[N-k] = \sum_{n=0}^{N-1} x[n] e^{j\frac{2\pi}{N}kn} = X^*[k] \tag{15}$$

Consequences of Symmetry:

- **DC Value:** $k = 0 \implies X[0] = \sum x[n]$. Purely real.
- **Nyquist Bin:** If N is even, $k = N/2$ is purely real since $X[N/2] = X^*[N/2]$.
- **One-sided Spectrum:** For N samples, bins from $k = 1$ to $N/2 - 1$ contain all unique frequency information.

3 Circular Shift of a Sequence

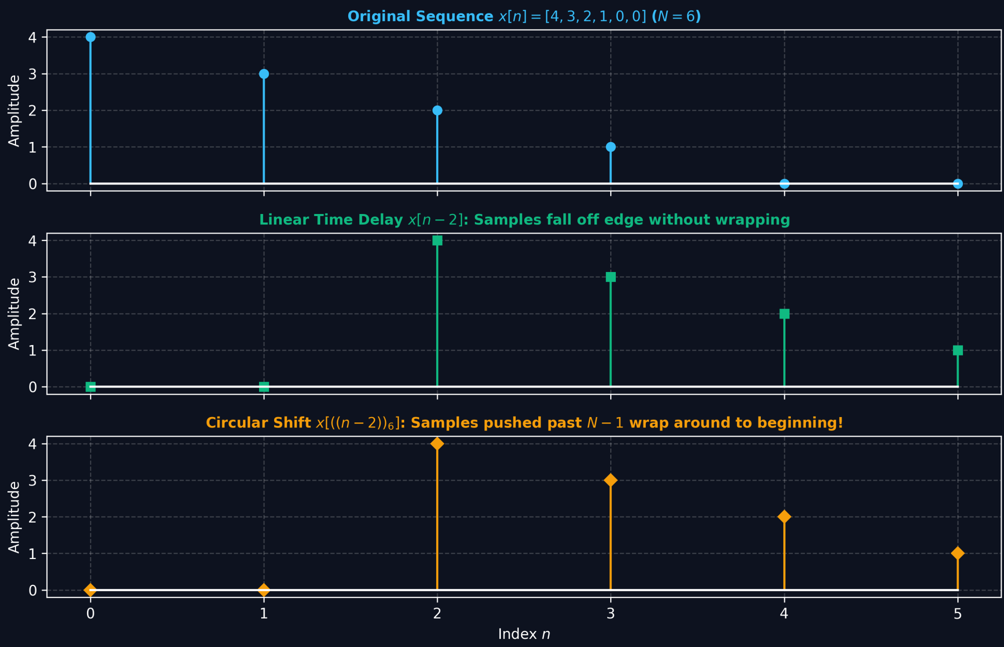

Figure 1: Linear Time Delay vs. Circular Modulo-N Shift on a Cylindrical Buffer.

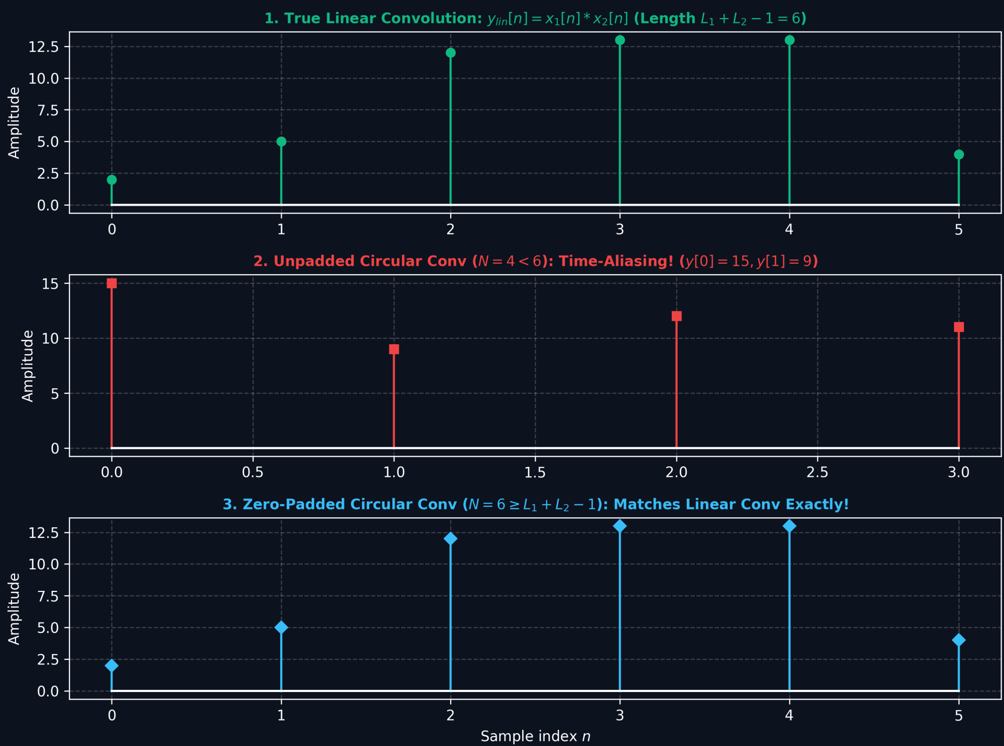

Figure 2: Linear vs. Circular Convolution: Aliasing Overlap ($N < L_1 + L_2 - 1$) vs. Zero-Padded Exact Recovery.

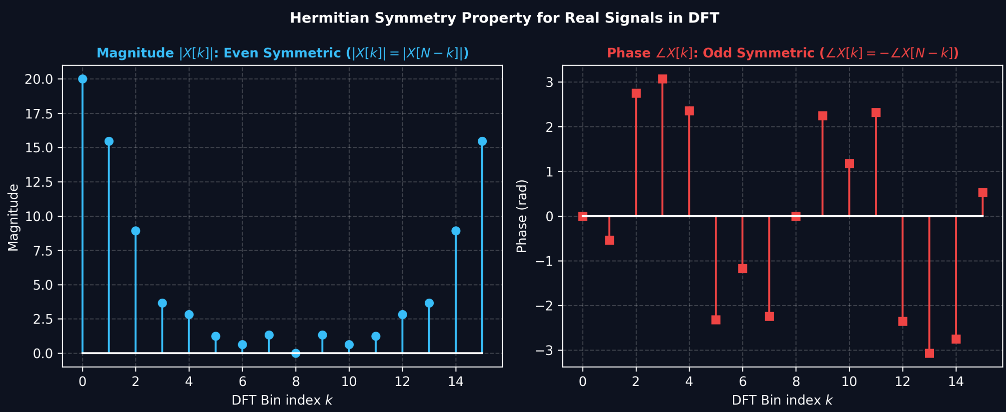

Figure 3: Hermitian Symmetry Properties and Parseval Energy Conservation for Real Sequences in the DFT.

3.1 Definition of Circular Shift

A circular shift of $x[n]$ by m samples is denoted using modulo indexing:

$$x_c[n] = x[((n-m))_N] \tag{16}$$

3.2 Circular Time Shift Theorem

$$x[((n-m))_N] \longleftrightarrow X[k]W_N^{km} \tag{17}$$

Proof:

$$X_c[k] = \sum_{n=0}^{N-1} x[((n-m))_N]W_N^{kn} \tag{18}$$

Let $l = n - m \implies n = l + m$:

$$X_c[k] = \sum_{l=0}^{N-1} x[((l))_N]W_N^{k(l+m)} \tag{19}$$

$$= W_N^{km} \sum_{l=0}^{N-1} x[l]W_N^{kl} \tag{20}$$

$$= W_N^{km} X[k] \tag{21}$$

4 Circular Convolution

4.1 Mathematical Definition

The circular convolution of $x_1[n]$ and $x_2[n]$ of length N is:

$$y[n] = x_1[n] \circledast_N x_2[n] = \sum_{m=0}^{N-1} x_1[m]x_2[((n-m))_N], \quad 0 \le n \le N-1 \tag{22}$$

4.2 Circular Convolution Theorem

$$y[n] = x_1[n] \circledast_N x_2[n] \longleftrightarrow Y[k] = X_1[k] \cdot X_2[k] \tag{23}$$

Proof:

$$Y[k] = \sum_{n=0}^{N-1} \left[\sum_{m=0}^{N-1} x_1[m] x_2[((n-m))_N] \right] W_N^{kn} \tag{24}$$
$$= \sum_{m=0}^{N-1} x_1[m] \left[\sum_{n=0}^{N-1} x_2[((n-m))_N] W_N^{kn} \right] \tag{25}$$
$$= \sum_{m=0}^{N-1} x_1[m] \left[X_2[k] W_N^{km} \right] \tag{26}$$
$$= X_2[k] \sum_{m=0}^{N-1} x_1[m] W_N^{km} \tag{27}$$
$$= X_2[k] \cdot X_1[k] \tag{28}$$

4.3 Circulant Matrix Representation

The operation $y = x_1 \circledast_N x_2$ is equivalent to $\mathbf{y} = \mathbf{H}\mathbf{x_1}$, where $\mathbf{H}$ is a circulant matrix:

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ \vdots \\ y[N-1] \end{bmatrix} = \begin{bmatrix} x_2[0] & x_2[N-1] & x_2[N-2] & \cdots & x_2[1] \\ x_2[1] & x_2[0] & x_2[N-1] & \cdots & x_2[2] \\ x_2[2] & x_2[1] & x_2[0] & \cdots & x_2[3] \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_2[N-1] & x_2[N-2] & x_2[N-3] & \cdots & x_2[0] \end{bmatrix} \begin{bmatrix} x_1[0] \\ x_1[1] \\ x_1[2] \\ \vdots \\ x_1[N-1] \end{bmatrix} \tag{29}$$

The eigenvalues of this circulant matrix are exactly the DFT coefficients $X_2[k]$.

4.4 Linear vs. Circular Convolution (Time Aliasing)

If linear convolution is $y_{lin}[n] = x_1[n] * x_2[n]$, circular convolution is:

$$y_{circ}[n] = \sum_{r=-\infty}^{\infty} y_{lin}[n - rN], \quad 0 \leq n \leq N-1 \tag{30}$$

Zero-Padding Rule: If $N \geq L_1 + L_2 - 1$, $y_{circ}[n] = y_{lin}[n]$. Otherwise, aliasing occurs.

4.5 Full Numerical Example: 4-Point Circular Convolution

Let $x_1[n] = \{1, 2, 3, 4\}$ and $x_2[n] = \{1, 0, 1, 0\}$. Find $y[n] = x_1[n] \circledast_4 x_2[n]$.

Method 1: Circulant Matrix Approach

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 4 \\ 6 \end{bmatrix} \tag{31}$$

So, $y[n] = \{4, 6, 4, 6\}$.

Method 2: DFT Multiplication Approach

- $X_1[k] = \{10, -2 + j2, -2, -2 - j2\}$
- $X_2[k] = \{2, 0, 2, 0\}$
- $Y[k] = X_1[k] X_2[k] = \{20, 0, -4, 0\}$
- $y[n] = \text{IDFT}(Y[k]) = \{4, 6, 4, 6\}$

5 Parseval's Theorem for DFT

Theorem:

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N}\sum_{k=0}^{N-1} |X[k]|^2 \tag{32}$$

Proof:

$$\begin{aligned}
E &= \sum_{n=0}^{N-1} x[n]x^*[n] && (33)\\
&= \sum_{n=0}^{N-1}\left(\frac{1}{N}\sum_{k=0}^{N-1} X[k]W_N^{-kn}\right)x^*[n] && (34)\\
&= \frac{1}{N}\sum_{k=0}^{N-1} X[k]\left(\sum_{n=0}^{N-1} x^*[n]W_N^{-kn}\right) && (35)
\end{aligned}$$

The inner sum is $X^*[k]$:

$$E = \frac{1}{N}\sum_{k=0}^{N-1} X[k]X^*[k] = \frac{1}{N}\sum_{k=0}^{N-1} |X[k]|^2 \tag{36}$$

6 Table of Key DFT Properties

Property	**Time Domain** $x[n]$	**Frequency Domain** $X[k]$
Linearity	$ax_1[n] + bx_2[n]$	$aX_1[k] + bX_2[k]$
Circular Time Shift	$x[((n-m))_N]$	$W_N^{km}X[k]$
Circular Freq Shift	$W_N^{-ln}x[n]$	$X[((k-l))_N]$
Conjugation	$x^*[n]$	$X^*[((-k))_N]$
Time Reversal	$x[((-n))_N]$	$X[((-k))_N]$
Circular Convolution	$x_1[n] \circledast_N x_2[n]$	$X_1[k]X_2[k]$
Multiplication	$x_1[n]x_2[n]$	$\frac{1}{N}X_1[k] \circledast_N X_2[k]$
Parseval's Relation	$\sum \|x[n]\|^2$	$\frac{1}{N}\sum \|X[k]\|^2$

7 Checkpoint Questions

Q1: If $x[n]$ is a 5-point sequence with DFT $X[k] = \{10, 2-j, 3+j2, 3-j2, 2+j\}$, find signal energy.
Answer: Using Parseval's theorem: Energy $= \frac{1}{5}(|10|^2+|2-j|^2+|3+j2|^2+|3-j2|^2+|2+j|^2) = \frac{1}{5}(100+5+13+13+5) = 27.2$.

Q2: For $h[n] = \{1, -1, 1\}$ and $x[n] = \{2, 1, 0, -1\}$, what is minimum DFT length to avoid aliasing?
Answer: $L_1 = 4$, $L_2 = 3$. Length $N \geq 4+3-1 = 6$.

Q3: Compute 3-point circular convolution of $x_1[n] = \{1, 2, 3\}$ and $x_2[n] = \{0, 1, 2\}$ via circulant matrix.
Answer:

$$\begin{bmatrix} 0 & 2 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \\ 4 \end{bmatrix}$$